DAY 1 · 19 OCTOBER 2026

Machine learning for medicine

Fundamentals, taught on real clinical data. Three hours of lecture, two hours of lab, one homework that becomes your project.

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How the three days run

DAY	LECTURE (3H)	PRACTICE (2H)	AFTER
19 Oct	ML fundamentals on medical data	Lab 1: breast cancer, end to end	HW1: Pima diabetes
20 Oct	ML in medical technology	Supervised project work	Finish project + slides
21 Oct	Defence only — 6 min talk, 4 min questions		Certificates

Everything runs on **scikit-learn and SciPy** . No GPUs, no deep-learning installs, no dataset you cannot load on the plane.

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Today

01	Why ML in medicine	15 min
02	What ML actually is	30 min
03	First model: linear regression	30 min
04	Classification	40 min
05	Generalisation and leakage	30 min
06	Evaluation in medicine	20 min

01

Why ML in medicine

Where it already shows up

01

Screening from images

Mammography and retinal screening: the model reads first, a clinician confirms. The gain is throughput, not genius.

02

Signals from devices

Arrhythmia detection on a watch or a patch ECG. The model runs on the device, on a battery, in milliseconds.

03

Risk scores in the ward

Sepsis, deterioration and readmission scores that sit inside the hospital record and fire an alert.

All three are the same maths you will write today. The difference is the evaluation and the consequences.

What it cannot do

It does not find causes. A feature with high importance is not a cause of the disease.

It does not generalise for free. A model trained in one hospital, on one scanner, on one population, is a model for that setting.

It does not replace judgement. It produces a number; the decision, the consent and the responsibility stay human.

It cannot fix the data. Missing values, label noise and selection bias survive every algorithm you throw at them.

What you will have built by Wednesday

3

Working notebooks: a diagnostic lab, a cleaned homework pipeline, a defended project.

2

Models compared honestly, with cross-validation and a stated uncertainty.

1

Operating point you chose yourself, and can justify to a clinician.

02

What ML actually is

Patients are rows, measurements are columns

PATIENT	RADIUS	TEXTURE	CONCAVITY	…	LABEL Y
001	17.99	10.38	0.300	…	malignant
002	13.54	14.36	0.045	…	benign
003	20.29	14.34	0.198	…	malignant

Breast Cancer Wisconsin: 569 patients, 30 features per patient, 212 malignant and 357 benign.

A model is a function with adjustable numbers

$$\hat{y} = f_\theta(x)$$

x is one patient's measurements. ŷ is the model's answer. θ are the parameters — the numbers learning changes.

$$\theta^* = \arg\min_\theta L(y, f_\theta(x))$$

Training is nothing more than searching for the θ that makes the loss L small. Inference is running f with those numbers on a new patient.

Four words you need by lunch

Supervised

Every row has a known answer. Learn the mapping from measurements to that answer.

Unsupervised

No answers. Find structure — groups of beats that look alike, for instance.

Regression

The answer is a number on a scale: disease progression in a year, blood glucose, length of stay.

Classification

The answer is a category: malignant or benign, positive or negative, which of five rhythms.

Predicting tumour size in millimetres: regression or classification?

And if the question becomes “is it larger than 20 mm?”

03

First model: linear regression

A weighted sum of the measurements

$$\hat{y} = w_1 x_1 + w_2 x_2 + \ldots + w_p x_p + b$$

One weight per measurement, one intercept. With ten features you are fitting eleven numbers.

Each weight reads as "the change in ŷ per unit of this measurement, holding the others fixed".
That reading is only honest when the features are not tangled together — which, in medicine, they always are.

How wrong is it?

$$\mathbf{MSE = (1/n)\,\Sigma\,(y_i - \hat{y}_i)^2}$$

Squared error: big mistakes hurt much more than small ones. In the units of y, squared.

$$\mathbf{R^2 = 1 - SS_{res} / SS_{tot}}$$

The share of variance explained. $R^2 = 0$ means you did no better than always predicting the mean. It can go negative.

Report the error in **clinical units** too. "MAE of 44 progression points" means something to a clinician; "$R^2 = 0.45$" does not.

Demo: predicting diabetes progression

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X, y = load_diabetes(
scaled=False , return_X_y=True, as_frame=True)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.2, random_state=42)

m = LinearRegression().fit(Xtr, ytr)
m.score(Xte, yte)
# R² on data the model never saw sorted(zip(X.columns, m.coef_), key=lambda t: -abs(t[1]))
```

442 patients, 10 measurements, target 25 to 346 : a one-year disease-progression score.

Why `scaled=False` matters

The default **load_diabetes()** returns features already centred and scaled and it uses the whole dataset, test rows included.

Coefficients then read in meaningless units, and the scaling itself has quietly seen your test set.

Load raw, and do every transformation **inside a Pipeline**. We will make this a rule in section 5.

04

Classification

Three models on the same data, and the one idea that separates a probability from a decision.

Logistic regression

$\sigma(z) = 1 / (1 + e^{-z})$

z is the same weighted sum as before. σ squeezes it into a probability between 0 and 1.

$L = -\Sigma\,[\, y \log \hat{p} + (1-y) \log(1-\hat{p}) \,]$

Log-loss. Being confidently wrong is expensive; being uncertain is merely mildly penalised.

The model outputs a **probability**. Turning it into "malignant" or "benign" needs a **threshold** and that threshold is a clinical choice, not a statistical one.

Two models with no equation to fit

k-nearest neighbours

Find the k most similar patients in the training set; vote. There is no training. The data is the model.

Scaling is not optional. One feature measured in thousands drowns every other distance. Small k memorises, large k blurs.

Decision tree

Repeatedly split on one measurement at a time: "concavity above 0.14?" Each leaf is a group of similar patients.

Readable, and unstable. A clinician can follow the path; change a few rows and the tree redraws itself. Depth controls everything.

Demo: three models, one split

```
from sklearn.datasets import load_breast_cancer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier

X, y = load_breast_cancer(return_X_y=True, as_frame=True)
y = 1 - y                          # make MALIGNANT the positive class models = {
    "logreg": make_pipeline(StandardScaler(), LogisticRegression(max_iter=5000)),
    "knn5":   make_pipeline(StandardScaler(), KNeighborsClassifier(5)),
    "tree4":  DecisionTreeClassifier(max_depth=4, random_state=42),
}
```

0 = malignant, 1 = benign

In **load_breast_cancer**, **target_names = ['malignant', 'benign']** . So the label 1 means *benign*.

Every scikit-learn metric treats **1 as the positive class** . A naive "sensitivity" is therefore sensitivity for healthy tissue.

Flip it on the first line of every notebook, and say out loud what "positive" means in your problem.

```
X, y = load_breast_cancer(return_X_y=True, as_frame=True) y = 1 - y   # malignant is now the positive class
```

05

Generalisation and leakage

A score on training data is not a score

Split first. 80 / 20, stratified, with a fixed random_state so your result is reproducible tomorrow.

Overfitting is a model learning the noise in your 455 patients instead of the biology behind them.

Bias and variance. Too simple and it is wrong everywhere; too flexible and it is wrong differently on every sample.

Touch the test set once , at the very end. Every decision you make by looking at it spends its honesty.

Five-fold cross-validation

Split the training data into five parts; train on four, score on the fifth; rotate.

Report **mean ± standard deviation** . One number without a spread is a claim you cannot defend.

Stratify so every fold keeps the class balance, critical with 270 or 583 patients.

0.972 ± 0.014

A defensible result. Two models separated by less than one standard deviation are not distinguishable — say so instead of picking a winner.

Three ways the test set leaks in

01

Scaling before splitting

The mean and standard deviation were computed over rows the model was not supposed to see.

02

The same patient twice

Two beats, two visits, two images from one person land on both sides. Split by patient, not by row.

03

Tuning on the test set

Choosing k, depth or a threshold by watching the test score. The number you then report is fiction.

Symptom: **excellent scores that collapse in the real world.** Leakage always flatters.

Pipeline makes leakage hard

```
from sklearn.pipeline import Pipeline
from sklearn.model_selection import cross_val_score, StratifiedKFold

pipe = Pipeline([
    ("scale", StandardScaler()),
    ("clf",   LogisticRegression(max_iter=5000)),
])

cv = StratifiedKFold(5, shuffle=True, random_state=42)
s = cross_val_score(pipe, Xtr, ytr, cv=cv, scoring="roc_auc")
print(f"{s.mean():.3f} ± {s.std():.3f}")
# scaler refit inside every fold
```

Every cleaning step including imputation, scaling, feature selection goes inside the pipeline. Nothing is fitted outside it.

Demo: watching a tree overfit

Fit a decision tree at every depth from 1 to 20. Record the **training** score and the **cross-validated** score.

Training accuracy climbs to a perfect 1.000 and stays there. That curve tells you nothing.

The CV curve rises, peaks around depth 3 to 5, and then sags. The peak is your model.

Where the two curves separate is exactly the memorisation you are paying for.

06

Evaluation in medicine

Accuracy is the wrong question. Here is the right one.

A model that detects nothing

63%

Accuracy of a classifier that answers "benign" for every single patient in the breast cancer data while missing all 212 malignant cases.

Always fit a **DummyClassifier** first. It is the number your real model has to beat.

The rarer the disease, the more flattering accuracy becomes, and the more useless.

For the rest of this course: **accuracy is never the primary metric** unless you can argue for it.

Four numbers, four consequences

	PREDICTED POSITIVE	PREDICTED NEGATIVE
ACTUALLY POSITIVE	**TP** Found the disease. Treatment starts.	**FN** Missed it. The patient goes home reassured.
ACTUALLY NEGATIVE	**FP** False alarm. Anxiety, biopsy, cost.	**TN** Correctly cleared.

The two errors are **not interchangeable**, and no single number tells you which one you are making.

Sensitivity and specificity

Sens = TP / (TP + FN)

Of the patients who have the disease, what fraction did the model catch? Screening lives and dies on this.

Spec = TN / (TN + FP)

Of the healthy patients, what fraction did the model leave alone? This is what keeps a screening programme affordable and humane.

Both are properties of the model **at one threshold** . Move the threshold and both change, in opposite directions.

The question the patient actually asks

PPV = (Sens · Prev) / (Sens · Prev + (1 – Spec) · (1 – Prev))

Given a positive result, what is the probability the patient really has the disease? Bayes' rule, written out.

Sensitivity and specificity are properties of the **test**. PPV is a property of the test **and the population you use it on** .

Same model. Two clinics.

SETTING	PREVALENCE	CALCULATION	PPV
Population screening	1%	0.009 / (0.009 + 0.099)	8.3%
Specialist clinic	30%	0.270 / (0.270 + 0.070)	79.4%

Sensitivity 0.90, specificity 0.90 in both rows. In the screening setting, **eleven out of twelve positives are false alarms** . The model did not change — the population did.

ROC and AUC

Sweep the threshold from 1 to 0 and plot **sensitivity against 1 – specificity** . That curve is the ROC.

AUC is the probability that a random sick patient is ranked above a random healthy one. 0.5 is a coin toss.

AUC is useful for **comparing models**. It never tells you which threshold to deploy.

With heavy imbalance, also look at **precision–recall AUC** — we come back to this tomorrow.

Demo: moving the threshold

```
p = pipe.predict_proba(Xte)[:, 1]        # probability of malignant for t in [0.2, 0.3, 0.5, 0.7]:
    tn, fp, fn, tp = confusion_matrix(yte, p >= t).ravel()
    print(t, "sens", tp/(tp+fn), "spec", tn/(tn+fp))
```

Lower threshold: catch more cancer, cause more false alarms. Higher threshold: the reverse. **There is no default value of 0.5 in medicine** — only a choice someone has to own.

07

Lab 1: breast cancer, end to end

lab1_breast_cancer.ipynb · two hours · you fill in the TODO cells

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The two hours

MINUTES	WHAT YOU DO
0–10	Environment check, load the data, flip the label
10–30	EDA: class balance, describe(), three histograms by class, correlation heatmap
30–45	Stratified 80/20 split, DummyClassifier baseline — and a discussion of why it scores 0.63
45–75	Logistic regression, k-NN for k in {1,3,5,11,21}, trees of depth 1 to 10, compared with 5-fold CV
75–100	Confusion matrix, then choose a threshold for sensitivity ≥ 0.95
100–112	Four of you present your operating point at the board
112–120	Form teams of three and write them down

You need sensitivity ≥ 0.95. Where do you choose the threshold?

```
from sklearn.model_selection import cross_val_predict

p_cv = cross_val_predict(pipe,
Xtr, ytr , cv=cv, method="predict_proba")[:, 1]
t = max(t for t in np.linspace(0, 1, 201) if sens_at(ytr, p_cv, t) >= 0.95)

pipe.fit(Xtr, ytr)
# now, and only now report(yte, pipe.predict_proba(Xte)[:, 1] >= t)
```

Choosing the threshold on the test set is **leakage** . Choose it with cross-validation on the training set; apply it to test exactly once.

Pima diabetes : 60 to 90 minutes

Count the zeros. Which are physiologically impossible ? glucose, blood pressure, skin thickness, insulin, BMI? Which are fine?

Replace impossible zeros with NaN , then impute the median inside a Pipeline. Compare CV ROC-AUC against the raw-zeros pipeline.

Compare logistic regression with a decision tree: 5-fold stratified CV, ROC-AUC, mean ± std.

Report sensitivity and specificity at threshold 0.5, and at the threshold that gives sensitivity ≥ 0.85, chosen on training data.

Written: a test with sensitivity 0.85 and specificity 0.80 compute PPV at prevalence 10% and at 34.9%. One sentence on what that means for general-population use.

Why this homework is the whole game

60%

of tomorrow's project is a clean HW1 notebook: load, clean, pipeline, cross-validate, choose a threshold, report a confusion matrix.

Write it so you can **swap the dataset on line one** and everything below still runs.

Functions, not copy-paste. You will reuse them under time pressure.

Deliverable: the notebook plus a five-sentence summary.

END OF DAY 1

Tomorrow: ML in medical technology

Ensembles on biomedical tables, interpretability, a raw ECG processed live, devices and hospital systems, and your project.

Bring HW1 finished · Teams of three, written down